

Bank Branch Management System Report

Bank Branch Management System Report

1. Introduction

The banking sector requires efficient systems to manage customer information, branch operations, accounts, and financial transactions. Manual handling of such operations can lead to delays, errors, and poor record management. To address these challenges, the Bank Branch Management System was developed as a database-driven solution that automates and organizes banking activities.

2. Problem Statement

Many small financial institutions and branch-based organizations face challenges such as poor customer record management, difficulty tracking transactions, data redundancy, and slow retrieval of information.

3. Objectives

Main Objective:

To design and implement a database system that manages bank branches, customers, accounts, and transactions effectively.

Specific Objectives:

- Store branch information
- Manage customer records
- Maintain account balances
- Record deposits and withdrawals
- Generate transaction reports

4. Technologies Used

- SQL
- MySQL
- GitHub
- VS Code

5. System Design

The system contains four tables:

- branches
- customers
- accounts
- transactions

6. Relationships

- One branch can have many customers
- One customer can own accounts
- One account can have many transactions

7. SQL Operations

- CREATE DATABASE
- CREATE TABLE

- INSERT INTO
- SELECT queries
- JOIN operations
- Foreign keys

8. Challenges Faced

- Understanding relationships
- Managing foreign keys
- Debugging SQL syntax

9. Achievements

The project successfully:

- Created a functional relational database
- Managed records effectively
- Generated transaction reports

10. Conclusion

The project demonstrates how relational databases can be used to manage banking operations efficiently. Future improvements may include graphical interfaces, authentication, and mobile banking integration.